

A Data-Driven Framework for Unaccompanied Children Care Pipeline Load and Flow Analytics

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Abstract—The management of unaccompanied children (UAC) requires continuous coordination between border processing and subsequent humanitarian care. This paper presents a data-driven analytical framework for examining reported UAC care-pipeline load and flow using a public U.S. Department of Health and Human Services (HHS) dataset. The study analyzes 720 valid reporting observations from January 12, 2023 to December 21, 2025 across six reported fields related to CBP custody, transfers, HHS care, and discharges. Two derived measures are introduced: Observed System Load, defined as the sum of reported children in CBP custody and HHS care, and Net Daily Intake, defined as transfers out of CBP custody minus HHS discharges. The analysis identifies substantial temporal variation, a peak observed system load of 11,762 children, and a marked reduction in reported load during 2025. Net Daily Intake was negative in 475 observations and positive in 238 observations. Data-quality validation identified 86 observations where reported transfers out of CBP custody exceeded reported CBP custody. These observations were retained and flagged rather than deleted. The results demonstrate how transparent descriptive analytics can be used to examine changes in care-pipeline load and flow while avoiding unsupported assumptions about physical facility capacity.

Index Terms—unaccompanied children, UAC program, data analytics, CBP custody, HHS care, pipeline analytics, operational load, time-series analysis

I. INTRODUCTION

The U.S. Unaccompanied Children (UAC) Program involves a sequence of custody and care stages beginning with apprehension and processing by U.S. Customs and Border Protection (CBP) and continuing through transfer to the Department of Health and Human Services (HHS) and eventual discharge. The HHS UAC dataset provides operational observations concerning children taken into CBP custody, children transferred out of CBP custody, children in HHS care, and children discharged from HHS care [1].

Operational datasets of this type provide an opportunity to examine changes in the reported care population over time without requiring confidential case-level information. However, the dataset contains both stock variables and flow variables. A stock variable represents the number of children reported in a particular stage at a given time, while a flow variable represents movement into or out of a stage. Treating these measures as interchangeable may lead to incorrect conclusions.

This study therefore separates two analytical concepts. The first is the observed system load, represented by the combined

TABLE I
FIELDS USED IN THE ANALYSIS

Dataset Field	Analytical Interpretation
Date	Reporting date
Children apprehended and placed in CBP custody	Reported daily apprehensions/placements
Children in CBP custody	Reported CBP custody stock
Children transferred out of CBP custody	Reported CBP-to-HHS transfer flow
Children in HHS Care	Reported HHS care stock
Children discharged from HHS Care	Reported HHS discharge flow

number of children reported in CBP custody and HHS care. The second is pipeline flow, represented by the difference between transfers out of CBP custody and HHS discharges.

The main contributions of this work are:

- development of transparent system-load and pipeline-flow indicators;
- validation of logical relationships among the reported variables;
- analysis of 720 valid reporting observations covering 2023–2025;
- comparison of annual care-load characteristics;
- identification of periods of positive and negative pipeline flow; and
- development of an interpretable analytical framework suitable for dashboard-based monitoring.

II. DATASET AND DATA PREPARATION

A. Dataset Description

The analysis uses the supplied extract of the HHS Unaccompanied Alien Children Program dataset. The dataset contains six principal fields used in this study. Their analytical interpretation is summarized in Table I.

The supplied CSV file contains 1,170 rows. After processing the reporting date and required numeric fields, 720 valid observations remained for the primary analysis. The valid reporting period extends from January 12, 2023 through December 21, 2025.

There are 720 unique valid reporting dates and no duplicate valid dates. However, the complete calendar interval contains 1,075 days. Therefore, 355 calendar dates do not contain a reported observation.

The missing dates are important because treating the dataset as a complete daily time series could produce misleading rolling statistics and visual continuity. No synthetic observations were inserted for the missing dates.

B. Data Preprocessing

The date field was converted to a standard datetime representation. The HHS-care field contained comma-formatted numerical values, which were converted to numeric values after removing comma separators.

The observations were sorted chronologically before calculating derived variables.

Missing calendar dates were preserved as missing rather than interpolated. This approach ensures that the analysis remains based on reported observations rather than estimated values.

III. DATA VALIDATION

Data validation was performed before calculating the analytical indicators.

Two logical checks were applied.

First, the number of children transferred out of CBP custody was compared with the reported number of children in CBP custody:

$$A_T = I(T_t > C_t), \quad (1)$$

where T_t represents children transferred out of CBP custody and C_t represents children in CBP custody.

Second, the number of children discharged from HHS care was compared with the reported HHS-care population:

$$A_D = I(D_t > H_t), \quad (2)$$

where D_t represents HHS discharges and H_t represents children in HHS care.

The analysis identified 86 observations in which reported transfers exceeded reported CBP custody. No observations were identified in which reported HHS discharges exceeded reported HHS care.

The 86 anomalous observations were not removed. Instead, they were retained and flagged. This approach preserves the source data and avoids making unsupported assumptions about reporting definitions, timing, or reconciliation procedures.

IV. ANALYTICAL METHODOLOGY

A. Observed System Load

The first derived metric measures the combined reported population in CBP custody and HHS care:

$$L_t = C_t + H_t \quad (3)$$

where:

- L_t = observed system load on reporting date t ,
- C_t = children in CBP custody,
- H_t = children in HHS care.

TABLE II
DESCRIPTIVE STATISTICS FOR VALID OBSERVATIONS

Measure	Mean	Median	Min	Max
Apprehended/placed	93.5	99.0	0	333
CBP custody	171.5	193.0	7	531
CBP transfers	128.7	157.0	0	440
HHS care	6061.3	6406.5	1972	11516
HHS discharges	173.4	181.0	0	505
Observed system load	6232.8	6633.5	2002	11762
Net Daily Intake	-44.7	-11.5	-465	206

This metric is referred to as *Observed System Load*. It should not be interpreted as facility utilization or capacity strain because the dataset does not provide information regarding available beds, staffing capacity, facility limits, or other operational capacity measures.

B. Net Daily Intake

The second derived metric measures the difference between reported transfers out of CBP custody and HHS discharges:

$$N_t = T_t - D_t \quad (4)$$

where:

- N_t = Net Daily Intake,
- T_t = children transferred out of CBP custody,
- D_t = children discharged from HHS care.

A positive value indicates that reported transfers exceeded HHS discharges for that observation. A negative value indicates that reported discharges exceeded transfers.

Net Daily Intake is interpreted as a pipeline flow indicator. It is not treated as an exact measure of total-system population change because changes in CBP custody and other processes are not represented by this equation.

C. Seven-Day Calendar Rolling Average

A seven-day calendar-based rolling average was calculated to identify short-term persistence in the reported flow:

$$\bar{N}_{7,t} = \frac{1}{|W_t|} \sum_{\tau \in W_t} N_\tau, \quad (5)$$

where W_t represents the available reported observations within the preceding seven calendar days.

Because 355 calendar dates are missing, this measure is specifically interpreted as a *seven-day calendar rolling average of reported Net Daily Intake*. Missing dates were not filled or interpolated.

V. RESULTS

A. Descriptive Statistics

The primary descriptive statistics are presented in Table II.

The mean observed system load was 6,232.8 children. HHS care accounted for the majority of this reported load, with an average of 6,061.3 children. The average reported CBP custody population was 171.5 children.

TABLE III
ANNUAL REPORTED LOAD CHARACTERISTICS

Year	Obs.	Avg HHS	Avg CBP	Avg Load	Peak Load
2023	230	8646.1	200.5	8846.5	11762
2024	251	7043.0	276.9	7319.9	10994
2025	239	2542.8	32.9	2575.7	6471

The maximum observed system load was 11,762 children on December 20, 2023. On this date, the reported HHS-care population was 11,516 and the CBP custody population was 246.

The minimum observed system load was 2,002 children on August 24, 2025.

B. Net Daily Intake Results

Net Daily Intake was negative in 475 of the 720 valid observations, positive in 238 observations, and zero in 7 observations.

Thus, approximately 66.0% of valid observations had negative Net Daily Intake, while approximately 33.1% had positive Net Daily Intake.

The maximum positive Net Daily Intake was +206 on February 12, 2024. On that date, 440 children were reported as transferred out of CBP custody and 234 children were discharged from HHS care.

The minimum Net Daily Intake was -465 on January 11, 2024. On this date, 11 children were reported as transferred out of CBP custody while 476 children were discharged from HHS care.

These observations demonstrate that daily pipeline flow can vary substantially even when the overall reported care population follows a different trajectory.

VI. ANNUAL ANALYSIS

Annual averages provide a useful high-level comparison of reported care load.

The results show a substantial reduction in reported care load over the study period.

The average HHS-care population declined from 8,646.1 children in 2023 to 7,043.0 in 2024 and 2,542.8 in 2025.

Similarly, the average observed system load declined from 8,846.5 children in 2023 to 7,319.9 in 2024 and 2,575.7 in 2025.

The reduction was particularly pronounced during the transition into 2025. Average HHS care was 6,418.7 in December 2024, 4,991.1 in January 2025, and 2,781.5 in February 2025.

The change in reported HHS care between December 2024 and February 2025 was approximately -56.7%.

This analysis identifies a substantial temporal change but does not establish its cause. The supplied dataset does not contain sufficient information to attribute the reduction to a specific policy, operational intervention, staffing change, or external event.

VII. RELATIONSHIP ANALYSIS

Correlation analysis was used to examine relationships among the reported variables.

The correlation between HHS care and HHS discharges was approximately 0.921, indicating a strong positive association. The correlation between HHS care and transfers out of CBP custody was approximately 0.714.

HHS care and Net Daily Intake showed a negative correlation of approximately -0.483.

The observed system load was highly correlated with HHS care, with a correlation coefficient of approximately 0.999. This is expected because HHS care represents the dominant component of the observed system-load calculation.

These correlations are descriptive. They do not establish causality or imply that changes in one variable directly caused changes in another.

VIII. DASHBOARD-ORIENTED ANALYTICAL FRAMEWORK

The analytical framework was implemented as an interactive dashboard to support exploratory analysis of the UAC care pipeline.

The dashboard provides:

- date-range filtering;
- Net Daily Intake filtering;
- peak observed system load;
- average observed system load;
- average HHS care load;
- average CBP custody load;
- average Net Daily Intake;
- peak Net Daily Intake;
- high positive and negative flow event tables; and
- data-validation warnings.

The dashboard uses the same derived indicators defined in this paper. In particular, the seven-day indicator uses a calendar-based rolling window rather than assuming that every observation represents a consecutive day.

The dashboard deliberately avoids using the term “capacity strain” as a direct measurement. The dataset contains no physical capacity variable. Therefore, the framework identifies observed load and pipeline pressure rather than actual capacity utilization.

IX. LIMITATIONS

Several limitations should be considered when interpreting the findings.

A. Incomplete Calendar Coverage

The dataset contains 720 valid reporting observations over a calendar interval of 1,075 days. Therefore, 355 calendar dates have no reported observation.

This prevents the analysis from being interpreted as a complete daily census. Rolling statistics are consequently based only on available reported observations within the relevant calendar window.

B. Reporting Anomalies

Eighty-six observations contain transfers out of CBP custody greater than the reported CBP custody population. The available dataset does not provide sufficient information to determine whether these differences result from reporting timing, definitions, or other operational factors.

The observations were therefore retained and flagged.

C. No Physical Capacity Data

The dataset does not contain facility capacity, available beds, staffing levels, utilization limits, or similar measures.

Consequently, high observed system load cannot be directly classified as capacity overload or capacity strain.

D. No Causal Inference

The study is observational and descriptive. Temporal changes and correlations should not be interpreted as causal effects of specific policies or operational interventions.

E. Stock-Flow Interpretation

Net Daily Intake is derived from two flow variables. It should not be interpreted as an exact change in total system population because changes in CBP custody and other processes are not represented.

X. FUTURE WORK

Future research can extend the proposed framework by incorporating additional variables that are not available in the supplied dataset.

Potential extensions include:

- facility-level capacity and bed availability;
- staffing levels and staff-to-child ratios;
- length of stay;
- medical and psychological service demand;
- placement and reunification outcomes;
- geographic or facility-level analysis;
- external policy and operational event data; and
- forecasting models for future care-pipeline load.

Such additional variables would allow future studies to move beyond observed load and examine utilization, operational constraints, and predictive planning more directly.

XI. CONCLUSION

This study presented a transparent data-driven framework for analyzing the UAC care pipeline using reported HHS and CBP operational data.

Across 720 valid observations from January 12, 2023 through December 21, 2025, the mean observed system load was 6,232.8 children and the maximum observed system load was 11,762 children. HHS care represented the dominant component of the reported system load.

Net Daily Intake was negative in approximately two-thirds of valid observations, indicating that reported HHS discharges exceeded transfers on most reporting observations. Nevertheless, substantial positive and negative flow events were observed.

A major reduction in reported care load occurred during 2025, with average HHS care falling from 8,646.1 children in 2023 to 2,542.8 children in 2025.

The analysis also identified 86 observations with a logical transfer-versus-custody inconsistency. Rather than deleting these observations, they were retained and explicitly flagged, supporting transparency and reproducibility.

The proposed framework demonstrates how descriptive analytics can provide meaningful insight into care-pipeline load and flow while maintaining appropriate analytical boundaries. In particular, the distinction between observed system load and physical capacity prevents unsupported conclusions from being drawn from population counts alone.

Future work incorporating facility capacity, staffing, length of stay, and outcome variables could provide a more comprehensive assessment of operational utilization and planning requirements.

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